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Experience

Full Stack Engineer

Hybrid

Cortucent Security

May 2024 - Present

- Most senior engineer; lead architecture and code review for healthcare revenue cycle (RCM) platforms.
- Build and maintain systems handling protected health information (PHI) under HIPAA and HITRUST.
- Implement HITRUST controls: encryption at rest and in transit, role-based access control, audit logging.
- Created the company's development standards for secure coding, code review, and releases.
- Develop full-stack applications for healthcare and other clients using FastAPI, Flask, React, and Node.js.
- Build AI systems for LLM document extraction, eligibility prediction, and data ETL pipelines.
- Modernized 1997-era legacy systems onto modern frameworks, increasing team productivity.
- Architect systems end to end (PostgreSQL, API, frontend) and deploy them with Azure and Docker.
- Built a Pub/Sub notification system on Twilio and Azure Communication Services.
- Implemented a logging and observability stack using Grafana OSS.
- Own the full design and implementation lifecycle of a messaging system integrating the Twilio API.

Education

Temple University

Philadelphia, PA

BS Computer Science

Aug 2021 - May 2025

- GPA: 3.74

Projects

Biogenie RAG

May 2025

- Built a full-stack RAG app for biological document search using Flask, React, and PGVector.
- Self-hosted embedding and generation models via Ollama, removing dependency on external LLM APIs.
- Implemented vector similarity search to retrieve contextually relevant documents at query time.

Sand Simulation

October 2024

- Built a real-time sand particle simulation in Zig using Raylib for hardware-accelerated graphics rendering.
- Implemented 2D physics using vector math to simulate realistic particle behavior and interactions.
- Demonstrated low-level systems programming by interfacing directly with a C library from Zig.

Technologies

Languages: Python, SQL, JavaScript, C, Go, Zig, Java, HTML/CSS, LaTeX

Technologies: Docker, PostgreSQL, FastAPI, React, Next.js, Claude Code, LLMs, GitHub Actions, Grafana, Pub/Sub, Azure, Flask, Node.js, Twilio, OpenAI API, PGVector, Git/GitHub, jQuery, PGAdmin

AI Skills

Development: AI-assisted SDLC (Claude Code, GitHub Copilot), prompt engineering, AI code review

Implementation: RAG pipelines, semantic embeddings, PGVector, Ollama (self-hosted LLMs), OpenAI API, LLM integration in production applications, agentic workflows and multi-step AI pipelines

Affiliations and Hobbies

Affiliations: ACM (Association for Computing Machinery), TUDevelopers

Hobbies: Photography, Guitar, Drawing, Collage, Making Websites for Friends, Compiling the Linux Kernel